Aishwarya Ainala

Backend Engineer | Software Development Engineer

IL, USA | +1 (773) 688-9003| aishwarya.a9000@gmail.com

SUMMARY

Backend Engineer with **4+ years of experience** building scalable, high-performance backend systems using **Java, Spring Boot, and microservices architecture**. Proven track record of improving system performance, reducing failures, and delivering production-grade APIs. Strong expertise in **Apache Kafka, AWS, relational and NoSQL databases**, and CI/CD automation. Experienced in developing **secure, fault-tolerant distributed systems**. Open to Backend Engineer / SDE roles with immediate availability.

TECHNICAL SKILLS

Programming Languages: Java (11 / 17), Python, SQL, Bash / Shell
Backend Development: Backend service development, RESTful API development, Microservices architecture, Event-driven systems, Asynchronous processing, Backend scalability, Performance optimization
Frameworks & Messaging: Spring Boot, Spring MVC, Spring Security, Hibernate / JPA, Apache Kafka
Databases & Caching: MySQL, PostgreSQL, MongoDB, Redis, Database indexing, Query optimization, Transaction management
Cloud & Infrastructure: AWS (EC2, S3, RDS, Lambda, IAM), Cloud-native backend development, High-availability systems, Fault-tolerant architectures, Auto-scaling concepts
DevOps & CI/CD: Docker, CI/CD pipelines, Jenkins, GitHub Actions, Build & deployment automation
System Design: Low-level system design, High-level system design, Distributed systems fundamentals, Load balancing, Service resilience patterns
Security: JWT authentication, OAuth2, Role-based access control (RBAC), Secure API design, OWASP Top 10
Testing & Quality: Unit testing (JUnit), Mockito, Integration testing, API testing (Postman), Code quality standards
Version Control & Process: Git, GitHub, Agile / Scrum, Code reviews

PROFESSIONAL EXPERIENCE

Capital One Financial, USA **Jan 2025 - Present**
Software Development Engineer (Backend)

- Improved system throughput by 38% by redesigning Java 17 + Spring Boot microservices for high-concurrency workloads.
- Built and secured REST APIs using Spring Security, JWT, and OAuth2, reducing authentication failures by 41%.
- Implemented Apache Kafka-based event-driven architecture, reducing request failures by 34% during peak traffic.
- Optimized PostgreSQL & MySQL queries using indexing and tuning, reducing API latency from 420ms to 290ms.
- Developed and deployed Dockerized services with GitHub Actions CI/CD pipelines, reducing rollback rates by 27%.
- Led backend initiatives across cross-functional teams, ensuring timely delivery of scalable and reliable microservices.
- Drove technical decisions for API design and microservices architecture, improving system scalability and maintainability.
- Mentored junior engineers through code reviews and design discussions, strengthening team capabilities in backend development.
- Lowered deployment rollback rates by 27% through Dockerized releases and automated GitHub Actions pipelines with pre-deployment validation checks.

CitiusTech, India **Apr 2020 - Jul 2023**
Backend Engineer / Software Development Engineer

- Developed Java 11 + Spring Boot backend services supporting 20K+ daily users.
- Improved data retrieval performance by 31% using SQL optimization, indexing, and schema tuning.
- Built event-driven systems using Apache Kafka, reducing downtime by 29%.
- Automated deployments using Jenkins and Docker, accelerating release cycles by 35%.
- Increased code quality with JUnit and Mockito testing, reducing production defects by 24%.
- Mentored and onboarded junior developers, improving team productivity and code quality standards.
- Contributed to system design discussions and guided solutions for complex backend issues.
- Collaborated with cross-functional teams to ensure smooth feature delivery across environments.
- Cut post-release defects by 24% through focused unit and integration testing using JUnit, Mockito, and API-level validation.

EDUCATION

Master's Degree | Illinois institute of Technology, Chicago, IL, USA
Bachelor's Degree | Vardhaman College of Engineering, Hyderabad, India